Three-dimensional data of wire-cut surface scans under the confocal microscope

Yuhang Lin[1,2,*], Heike Hofmann[2,3], Curtis Mosher[4], Eden Amin[5], Jeff Salyards[2], Alicia Carriquiry[1,2]

[1]Department of Statistics, Iowa State University, USA
[2]Center for Statistics and Applications in Forensic Evidence, Iowa State University, USA
[3]Department of Statistics, University of Nebraska-Lincoln, USA
[4]Roy J. Carver High Resolution Microscopy Facility, Iowa State University, USA
[5]Forensic Science Institute, University of Central Oklahoma, USA
[*]Corresponding author: Yuhang Lin (yhlin@iastate.edu)

Abstract

Wire-cut striation marks provide evidence for forensic source identification, but publicly available datasets and reproducible quantitative workflows for analyzing these marks are limited. We present a dataset of 120 three-dimensional topographic scans in `x3p` format from aluminum wires cut with five KWS-105 wire cutters. The study contains six scans—two replicates at each of three cutting locations—for each of the 20 tool-edge combinations. We provide additional scan metadata, manually extracted profiles, processed signals, cross-correlation function (CCF) values for pairwise aligned signals, images, and analysis code. The accompanying workflow covers scan inspection, profile and signal extraction, alignment, pairwise similarity visualization, CCF distributions for same-source and different-source comparisons, and variability assessment across replicates, stagings, and acquisitions. The dataset and workflow support the development, evaluation, and reproducible comparison of quantitative methods for wire-cut evidence.

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Contents

Background & Summary

In forensic analysis, marks left at crime scenes play an important role, and forensic examiners are particularly interested in identifying their origin, a process known as the Source Identification Problem in Forensic Science[1]. The forensic science community focuses on two distinct problems: the **specific source problem**, in which examiners aim to determine whether a mark was left by a particular tool, and the **common source problem**, in which examiners aim to determine whether two marks were left by the same tool. Currently, accepted practice for both problems relies on visual inspection of physical items under a comparison microscope. Examiners then report their conclusions according to the Theory of Identification[2], developed by the Association of Firearm and Toolmark Examiners (AFTE), using one of three categories: *identification*, in which the marks are believed to have been made by the same source; *elimination*, in which the marks are believed to have been made by different tools; and *inconclusive*, in which the similarities or dissimilarities between marks are insufficient to support either identification or elimination. This process has been criticized by the National Research Council (NRC)[3] and the President's Council of Advisors on Science and Technology (PCAST)[4] for its lack of objectivity and absence of quantifiable measurements, such as error rates.

In discussing error rates, Biedermann[5] differentiates between internal and external perspectives. The external perspective allows general statements based on black-box studies that relate examiners' conclusions to ground truth; however, it does not consider the evidence in a particular case. Introducing an internal perspective requires evaluating similarity between pairs of specimens with known ground truth. Algorithms allow these similarities to be quantified objectively and reproducibly.

Advances in microscopy have made it possible to move from optical two-dimensional imaging to digital three-dimensional topographic scanning, allowing algorithmic assessment of the similarity between pairs of forensic pattern evidence[6]. This transition has already supported practical advances in bullet and cartridge comparisons, including visual inspection of bullet line scans from three-dimensional topographic images[7], the use of features extracted by fast Fourier transform (FFT) as the basis of an artificial intelligence system[8], an automated pilot study based on consecutively matching striae[9] (a feature also used by some laboratories in casework), and the incorporation of three-dimensional imaging into routine casework at the Alabama Department of Forensic Sciences (ADFS)[10].

When cut wires are found at a crime scene, understanding the characteristics of marks left by bladed tools becomes essential. When a blade cuts a wire, it leaves striations on the surface, as shown in Figure 1. These striations provide a basis for similarity assessment and source identification. Similar scans have supported the development of quantitative methods for analyzing firearms and other toolmark evidence. For example, studies of screwdriver striations have examined key factors affecting comparisons, including angles of attack[11], rotational axes[12], and cutting directions[13]. Foundational research on forensic toolmarks has also emphasized data collection and analysis, including the collection and distribution of datasets for bullet and toolmark analysis[14,15], numerical methods for improving accuracy and consistency[16,17], and techniques for quantifying similarity[18,19].

Data collected at crime scenes lack the known ground truth needed to estimate error rates; therefore, controlled studies are needed. In this study, we provide a publicly available dataset of wire-cut scans together with a systematic quantitative processing and comparison workflow. The shared materials include raw scans, metadata, manually extracted profiles, processed signals, pairwise cross-correlation function (CCF) values, derived images, and analysis code.

The Methods section describes sample collection, scanning, profile extraction, signal processing, and CCF-based alignment. Data Records describes the shared files and repository organization. Data Overview presents descriptive summaries of pairwise CCF values. Technical Validation assesses signal-processing and alignment behavior and variability across replicates, stagings,

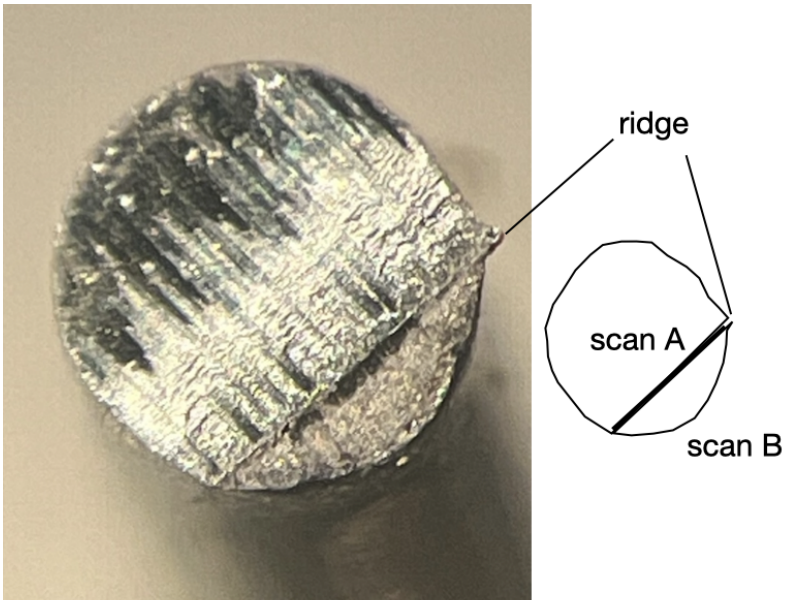

Figure 1. Microscopic wire-cut striations. Close-up of striations left by a blade on the cut end of a wire; the schematic identifies the ridge separating the two roof-like surfaces scanned separately as scans A and B.

and acquisitions. Usage Notes provides practical guidance for working with the shared files, and Code Availability identifies the custom processing and validation code. The dataset and workflow may support future quantitative research on wire-cut evidence.

Data Acquisition and Processing

Five Kaiweets KWS-105 multipurpose tools with 0.5-inch carbon-steel cutting blades were used to cut 16-gauge anodized pure-aluminum wire.
While aluminum is not commonly used for electric wiring because of its poor conductivity, it was selected for its nontoxicity and suitability for receiving cutting marks and retaining them during subsequent handling. Pure aluminum has a Brinell hardness of approximately 15 HB, which places it between lead (approximately 5 HB) and copper (approximately 35 HB).

Scan acquisition

The blades of each of the five wire cutters were labeled A, B, C, and D (see Figure 2 (a)). B opposes A; the reverse sides are labeled D for A and C for B. Three locations on each blade were marked (inner, middle, and outer), as shown in Figure 2 (a). Each four-inch wire segment was cut in half, creating two cut sides, AB and CD. The cut surface of the wire is not flat; under the microscope, it reveals a roof-like structure with a ridge separating the markings made by each blade (see Figure 1). The two roof sides of each cut surface were scanned separately, yielding scans A and B from cut side AB and scans C and D from cut side CD. As an example, the four scans resulting from cut 1 by tool 1 are shown in Figure 2 (c).
The cut surfaces were scanned at 20× magnification using a Sensofar confocal microscope, shown in Figure 2 (b). The scans were exported in `x3p` format (ISO 25178-72:2017/Amd 1:2020) with a lateral resolution of 0.645 × 0.645 micrometers per pixel and a vertical resolution of 0.1 micrometer.
The complete design therefore consists of 5 tools × 3 locations × 2 replicates × 4 cut surfaces = 120 scans.
The naming of each piece included the specific tool, its edge, location, and replicate according

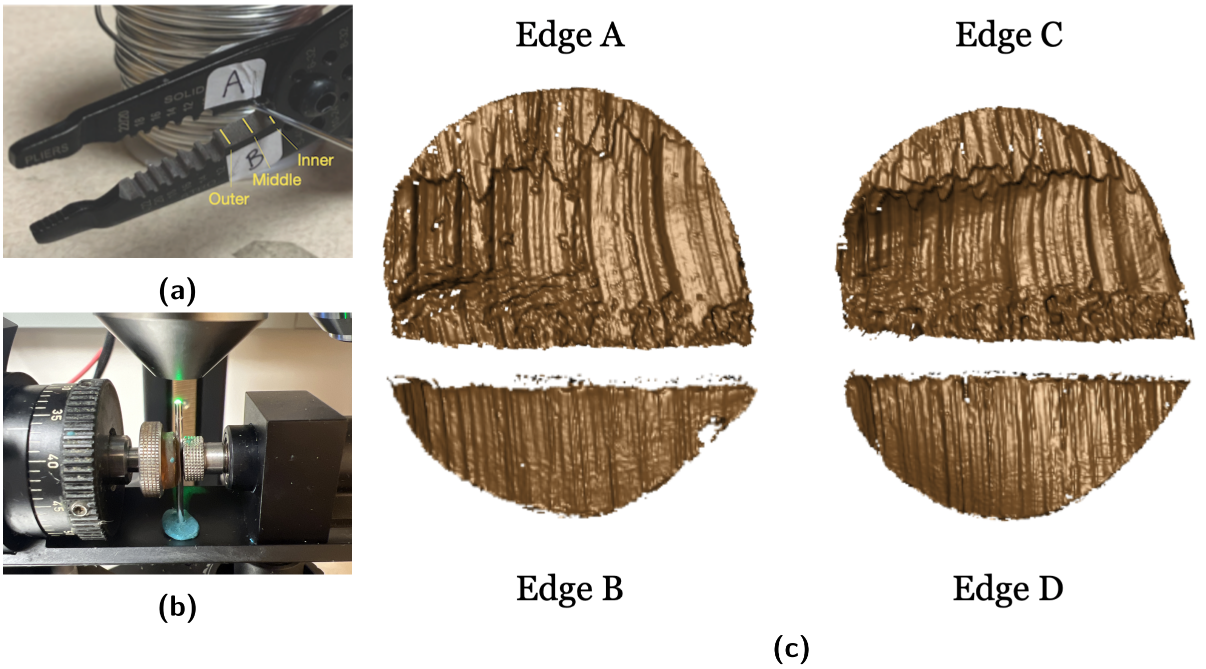

Figure 2. Wire-cut sampling and scanning workflow. (a) A Kaiweets KWS-105 wire cutter with inner, middle, and outer locations marked. (b) A confocal microscope used to scan wire-cut surfaces. (c) Four scans of cutting marks labeled according to the corresponding cutting edges A, B, C, and D.

to the following convention:

```
T__W-L_-R_
 ||   |  |
 ||   |  +-- Replicate: 1 or 2
 ||   +----- Location: I (inner), M (middle), or O (outer)
 |+--------- Edge: A, B, C, or D
 +---------- Tool: 1, 2, 3, 4, or 5
```

For example, T1AW-LI-R1 denotes the first replicate of a piece cut with edge A of tool 1 at the inner location.

To minimize errors during data collection, two team members cut and labeled the wire pieces together. One team member scanned the wire cuts and assigned names to the scans as described.

Signal extraction

Signals were extracted from each surface scan using a two-step procedure. First, a representative profile was identified along a line across the surface perpendicular to the dominant striation direction. This line was chosen manually to avoid scanning artifacts (e.g., holes) and cutting artifacts (e.g., rips caused by rolling the wire). Along this line, surface values were assigned at equally spaced positions using the observed surface value at the nearest position on the surface grid.

In a second step, we extract the scan's signal by removing trends and outliers from the surface values of the profile with the help of two LOWESS filters with bandwidths of $400\mu m$ and $40\mu m$, respectively[20]. All smoothers are susceptible to boundary issues, particularly when outliers are present. Steep changes at the end of a signal have a high potential to attract another signal during alignment, causing misalignments. To avoid potential artificial "hooks" at either end of a signal, two-sample t-tests were applied to the surface values in the outermost 2.5% intervals along the horizontal axis. Specifically, the summary statistics (sample size, mean, and

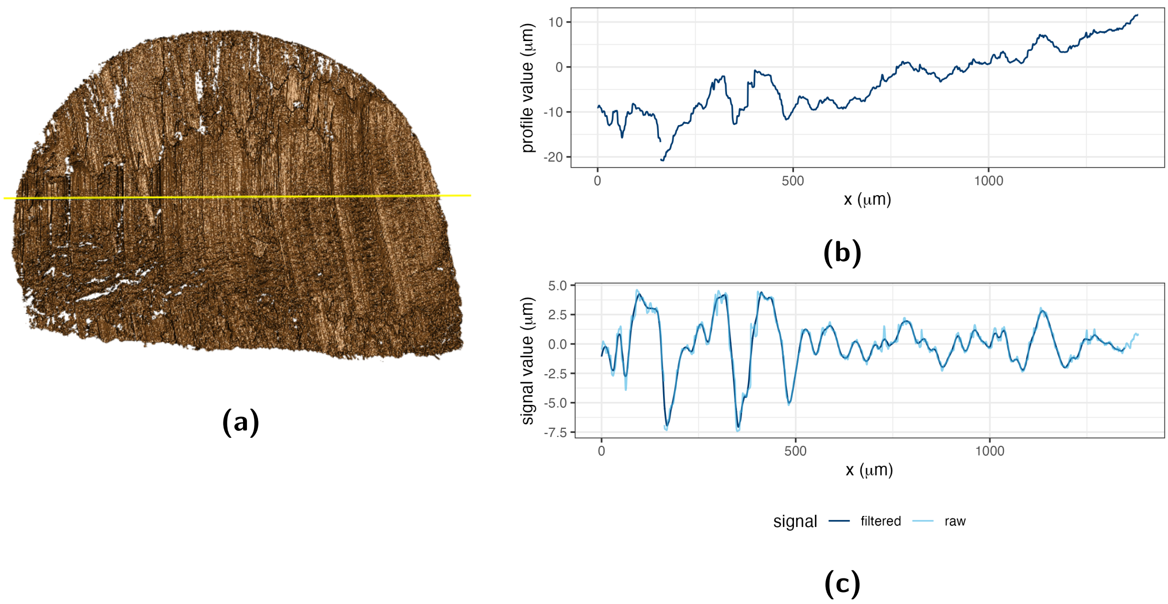

Figure 3. Profile and signal extraction. (a) A yellow profile line drawn across the scan striations. (b) The profile function extracted along the line in (a). (c) Raw and filtered signals obtained from the profile function in (b).

standard deviation) of the signal values within the intervals $[0, 2.5)\%$, $[2.5, 5)\%$, $(95, 97.5]\%$, and $(97.5, 100]\%$ were calculated. At each end, the test compared the outermost 2.5% interval with the adjacent interval. The signal values in the outermost interval were retained only when the corresponding p-value was above 0.01, indicating no significant shift.

The widths of these intervals correspond to the choice of bandwidths and the length of the signal. The interval widths were determined using cross-validation.

The resulting signals are used for subsequent comparisons with other signals.

Figure 3 shows a scan on the left (a) with the location of the manually chosen profile line in yellow. The corresponding profile and signal are plotted in (b) and (c).

One profile for each scan is included in folder `profiles/` as a csv file in the repository using the same naming convention as the scans, with the precursor of `profiles-`. More details on the exact format of these files are provided below.

A single file containing all 120 signals is available in `data-derived/wire-signals.csv.zip`.

Pairwise alignment of signals

Before signals from different scans can be compared, they need to be aligned. For each pair of signals, the value of the cross-correlation function (CCF) is calculated across relative shifts. The shift corresponding to the maximum CCF is retained as the alignment position, while the maximized cross-correlation function, CCF_{max}, gives a measure of the similarity of the signals, with a value of 1 indicating complete similarity (apart from linear scaling) and a value of 0 indicating (statistical) independence.

Figure 4 shows all signals corresponding to cuts 1 and 2 made with tool 1 at the location closest to the pivot area (inner location). Each of the four edges corresponds to one row in the figure. The lines in the first two columns correspond to the extracted signals for cuts 1 and 2, respectively. The third column shows the two signals aligned at the location of the maximal CCF. The cross-correlation value for each pair is included as part of the title of the chart.

The file `data-derived/wire_pairwise_ccf.csv` contains the CCF similarity scores of all pairwise aligned signals.

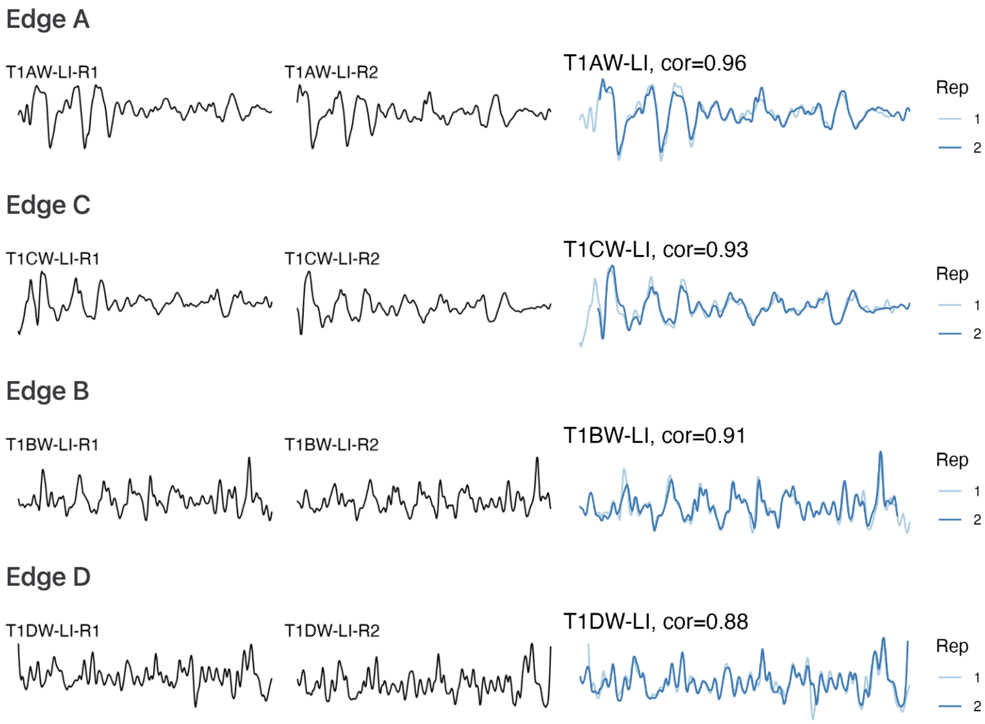

Figure 4. Same-source signal alignment. Rows correspond to cut surfaces A, C, B, and D. The first two columns show signals extracted from scans corresponding to the replicate cuts of tool 1 at location I; the third column overlays the aligned signals and reports their CCF values.

Data Records

The repository record on Iowa State University DataShare [DATASET CITATION] at `https://doi.org/[DOI]` contains the x3p scans, metadata, extracted profiles, processed signals, pairwise CCF values, and derived images. [Agent: Replace the dataset citation and DOI placeholders. For first-round review, the destination must permit anonymous download; from the second round onward, it must be a public formal repository record. Add the full dataset reference to References.]

The repository contents, formats, and folder organization are summarized in Table 1. [Agent: The `variability-assessment/` files have not yet been updated to the current naming conventions. Recheck the corresponding inventory entry and manuscript descriptions after those files are revised.]

A data dictionary defining non-obvious columns in `meta.csv`, the profile and signal files, and the pairwise-CCF file is provided in [repository documentation path]. The dataset is available under the [license] license. [Agent: Verify Table 1 against the deposited record and replace the documentation-path and data-license placeholders. For this non-sensitive original dataset, use CC0 or CC BY rather than a license containing NC or SA restrictions.]

[Heike: We need to include a descriptor table for each of the csv files; I started with meta.csv - the formatting is still horrific]

The `meta.csv` file contains one record for each of the 120 scans, with eight variables described in Table 2.

The `profiles/` folder contains 120 CSV files, one manually extracted profile per scan, with one sampled point per row. Their four variables are described in Table 3 in CSV column order.

The `data-derived/wire-signals.csv.zip` archive contains `data-derived/wire-signals.csv`, containing signals for all 120 scans. Table 4 describes its 12 variables in CSV column order.

Table 1. Structure of available data and files.

Data type	Description	Section
Raw data		
`scans/`	folder containing 120 topographic 3D scans corresponding to 30 aluminum wire cuts (`x3p` format)	Scan acquisition
`meta.csv`	metadata for each scan: identifier, file path, tool, edge, location, replicate, and pixel dimensions (CSV format)	Scan acquisition
Manual derivatives		
`profiles/`	folder containing one manually extracted profile per scan (CSV format)	Signal extraction
Computational derivatives in the folder 'data-derived/'		
`wire-signals.csv.zip`	profile values and derived signals for all 120 scans (CSV format, in `wire-signals.csv.zip`)	Signal extraction
`wire_pairwise_ccf.csv`	CCF values for all pairwise aligned signals (CSV format)	Pairwise alignment of signals
Image files		
`pngs/`	folder containing images of 3D wire-cut scans (PNG format)	Scan acquisition
`profile-images/`	folder containing images of profiles extracted from wire cuts (PNG format)	Signal extraction
Technical validation		
`variability-assessment/`	folder containing technical-validation documents, pending an update to the current naming conventions	Technical Validation

Table 2. Structure of `meta.csv`.

Variable	Type	Description
`ID`	text	Unique scan identifier defined by the naming convention in Scan acquisition
`x3p_path`	text	Path to the `x3p` scan relative to the data repository root
`tool`	integer	Cutting tool, one of 1 through 5
`edge`	character	Cutting edge, one of A, B, C, or D
`location`	character	Cutting location: I (inner), M (middle), or O (outer)
`replicate`	integer	Replicate cut, either 1 or 2
`height`	integer	Height of the scan in pixels
`width`	integer	Width of the scan in pixels

The `data-derived/wire_pairwise_ccf.csv` file contains all 14,400 ordered pairs of the 120 signals. Table 5 describes its 9 variables in CSV column order.

Table 3. Structure of the `profiles-*.csv` files.

Variable	Type	Description
`ID`	text	Source scan identifier, matching `ID` in `meta.csv`; constant within each file
`x`	numeric	Horizontal coordinate along the profile in micrometers, starting at zero in increments of 0.645
`y`	numeric	`y` coordinate, constant and set to zero
`value`	numeric	Observed surface height in micrometers; `NA` denotes a missing value

Table 4. Structure of the signal CSV in `wire-signals.csv.zip`.

Variable	Type	Description
`ID`	text	Source scan identifier, matching `ID` in `meta.csv` and the profile files
`tool`	integer	Cutting tool, one of 1 through 5
`edge`	character	Cutting edge, one of A, B, C, or D
`tooledge`	text	Combined tool and edge label, for example, `T1AW`
`location`	text	Cutting location: I (inner), M (middle), or O (outer)
`replicate`	integer	Replicate cut, either 1 or 2
`x`	numeric	Horizontal coordinate along the profile in micrometers, starting at zero in increments of 0.645
`y`	numeric	`y` coordinate, constant and set to zero
`value`	numeric	Observed surface height in micrometers
`detrend`	numeric	Surface height after subtracting the fitted linear trend in `value` versus `x`, in micrometers
`raw_sig`	numeric	Residual in micrometers after removing the 400-micrometer LOWESS trend from `detrend`, before final smoothing and boundary exclusion
`sig`	numeric	Signal after 40-micrometer smoothing of `raw_sig` and boundary exclusion, in micrometers

Table 5. Structure of the `wire_pairwise_ccf.csv` file.

Variable	Type	Description
`ID1`	text	Scan identifier for the first signal, matching `ID` in `meta.csv`
`ID2`	text	Scan identifier for the second signal, matching `ID` in `meta.csv`
`ccf`	numeric	Maximized cross-correlation function value for the aligned signals
`lag`	integer	Relative shift at which `ccf` is maximized, in profile-sample increments
`same_tool`	logical	Whether both signals were produced by the same tool
`same_edge`	logical	Whether both signals were produced by the same tool and edge
`same_location`	logical	Whether both signals were produced by the same tool, edge, and location
`same_source`	logical	Whether `same_location` is true; replicates may differ
`edge_type`	text	Edge-length pairing: Long, Short, or Mixed

Data Overview

As descriptive summaries of comparison behavior across the dataset, Figure 5 displays pairwise CCF values organized by tool, cut surface, blade location, and replicate. The main diagonal contains self-comparisons, whereas the off-diagonal cells compare distinct signals. Figure 6 displays the CCF distributions for same-source and different-source comparisons.

Figure 5. Pairwise CCF tile map. Rows and columns group scans by tool, cut surface, blade location, and replicate. Orange cells indicate larger CCF values and gray cells indicate smaller values; the inset enlarges the boxed edge-to-edge comparison block.

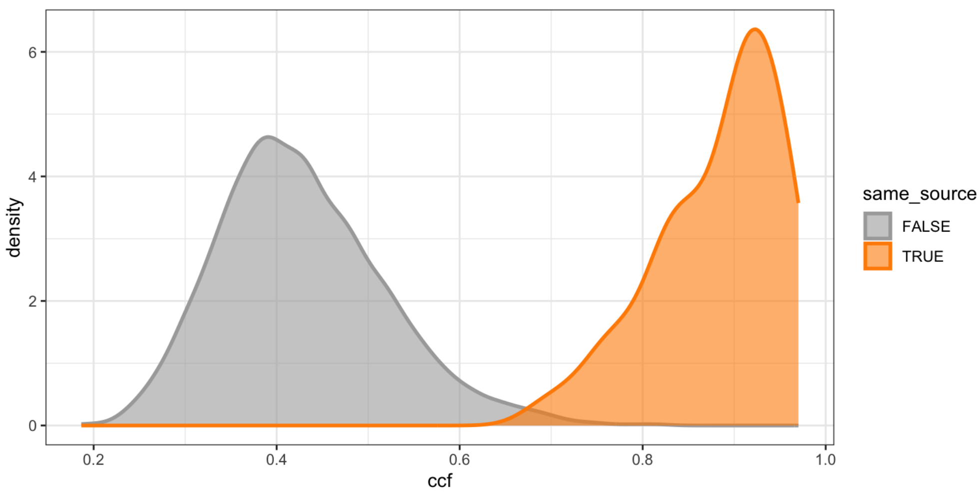

Figure 6. CCF density distributions. Kernel density estimates are shown for different-source comparisons in gray and same-source comparisons in orange.

Technical Validation

Technical validation was conducted in two areas: (1) signal processing and alignment and (2) variability assessment.
Scan processing and signal alignment were validated as follows. High CCF values were expected between signals from the same source and low values between signals from different sources. Figure 4, shown earlier in Pairwise alignment of signals, meets the expected pattern for same-source signals. As an example of a different-source comparison, T1AW-LI-R1 and T2AW-LI-R1 were selected. Figure 7 (a) and (b) show the selected scans. Their signal alignment, shown in Figure 7 (c), has a CCF value of 0.2, which is as low as expected. For the following summaries, edges A and C were categorized as long, edges B and D as short, and comparisons pairing one long edge with one short edge as mixed. The resulting CCFs for all pairwise comparisons were also summarized using boxplots and receiver operating characteristic (ROC) curves. Figure 8 groups CCF distributions by different edge, different location, different tool, and same source, with the paired edges identified on the horizontal axis. Figure 9 shows classification performance separately for long and short edge classes. The ROC curves plot sensitivity, or the true positive rate, against the false positive rate (FPR; 1 - specificity). The curves are close to the upper-left corner, indicating strong classification performance. For the combined comparison set, a threshold of 0.589 controls the FPR below 0.05 with a false negative rate (FNR) of 0, whereas a threshold of 0.658 controls the FPR below 0.01 with an FNR of 0.02. These results are consistent with the expected pattern and validate the pipeline for processing and aligning signals.

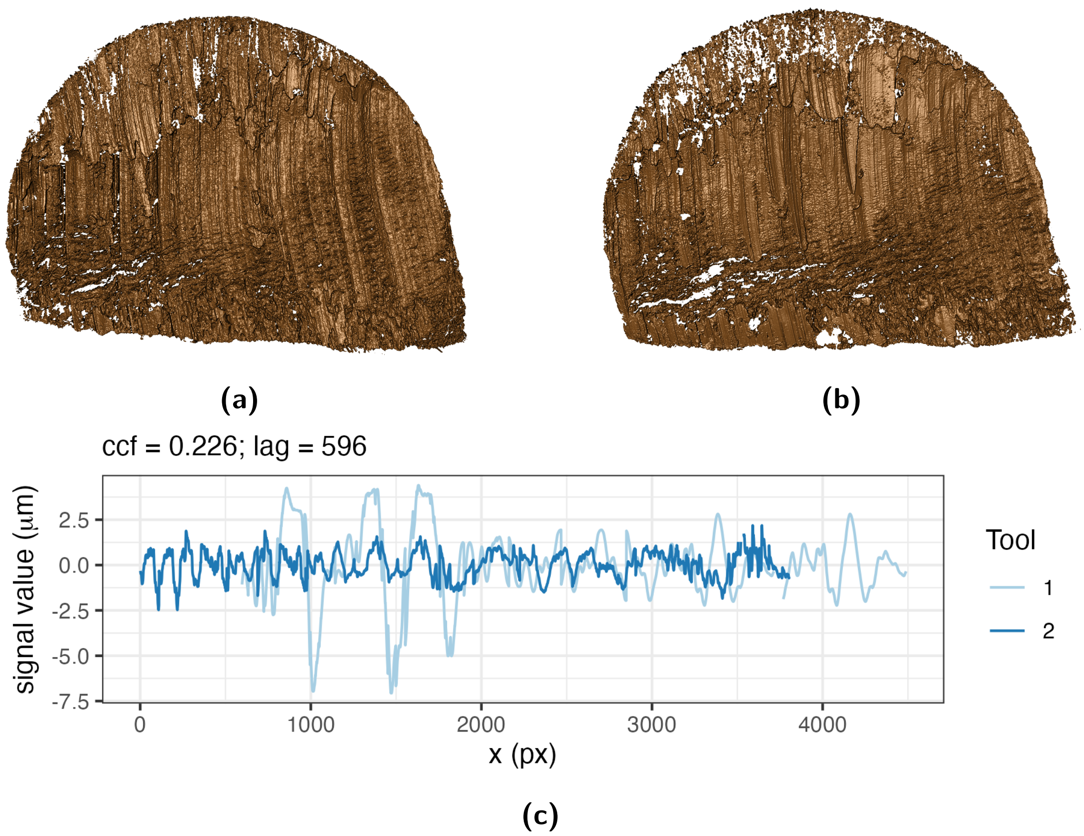

Figure 7. Different-source scan and signal comparison. (a) Scan T1AW-LI-R1 cut by tool 1. (b) Scan T2AW-LI-R1 cut by tool 2. (c) Alignment of signals from T1AW-LI-R1 and T2AW-LI-R1.

The second validation assessed variability in the pipeline across replicates, stagings, and acquisitions. Scans from tool 2 at the middle location were selected for this assessment. Replicate variability was assessed using two different cuts at the same edge and position. The compari-

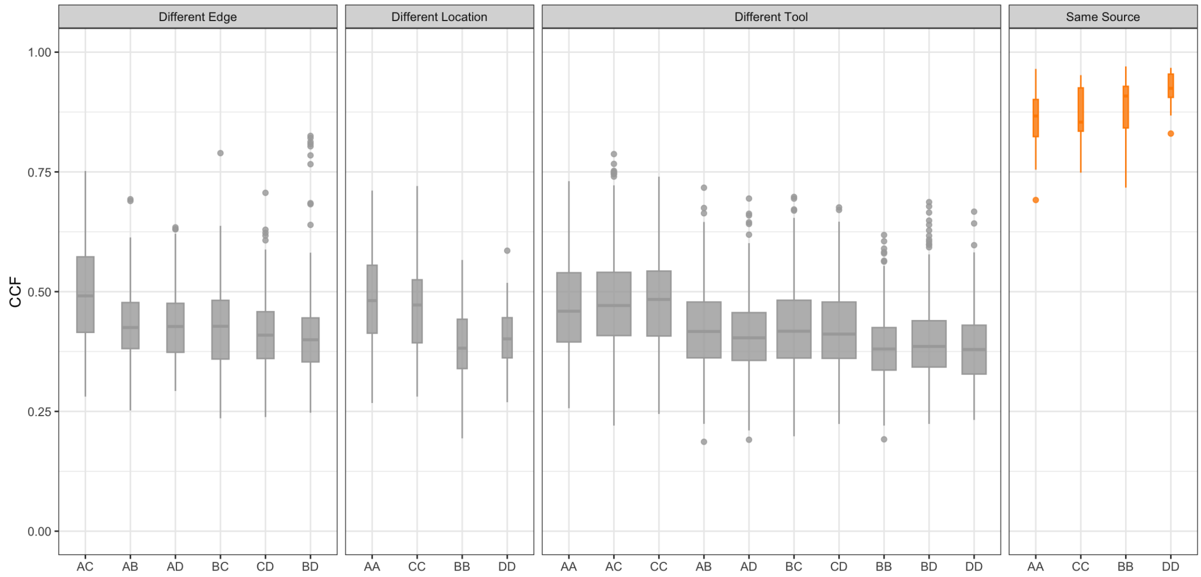

Figure 8. CCF distributions across comparison classes. Gray boxplots show different-source comparisons grouped by different edge, location, or tool; orange boxplots show same-source comparisons. Horizontal-axis labels identify the paired edges.

son in Figure 4 in Pairwise alignment of signals illustrates this assessment. To assess staging variability, replicate 2 was rescanned under the confocal microscope, and the staging label S was introduced into the naming protocol. The S1 designation is omitted from labels, so R2 is equivalent to R2-S1. For all four edges of tool 2 at the middle location, three stagings (R2, R2-S2, and R2-S3) were created, pairwise CCFs between stagings were computed, and their mean and standard deviation (SD) were calculated.

Within each staging, the effect of different acquisitions was further examined by keeping the samples on the confocal microscope while varying the lighting conditions. A was introduced as the acquisition label. For edge A at staging 3, four acquisitions were performed, and the A1 designation was omitted from the first acquisition label. Therefore, R2-S3 is equivalent to R2-S3-A1. For the other three edges, two acquisitions were conducted at staging 3. These acquisitions are denoted by R2-S3 and R2-S3-A2. Within staging 3, pairwise CCFs between acquisitions were computed, and their mean and standard deviation were calculated. Variability between distinct stagings was assessed separately through the staging comparisons described above.

With this setup, CCF values were expected to become more consistent from replicate to staging to acquisition as the scanning conditions became progressively more stable. Consequently, variability was expected to decrease along this sequence. Replicates, stagings, and acquisitions were also compared, with smaller differences expected between acquisitions and stagings than between stagings and replicates. The results were visualized as scatter plots. Figure 10 shows the scatter plots and intervals extending two standard deviations above and below the mean. Higher orange mean lines indicate an increase in the mean CCF, and narrower shaded intervals indicate a decrease in the standard deviation. The detailed numerical results, including means, standard deviations, and differences between means, are shown in Table 6 and confirm the expected pattern.

[Agent: The manuscript contains 11 figures. Scientific Data recommends no more than eight but does not mandate this limit. Confirm with the coauthors or professor whether any logically connected pairs should be combined for revised submission; do not remove a figure or its discussion without an author decision.]

The variability assessment does not include variation in CCF values caused by differences in

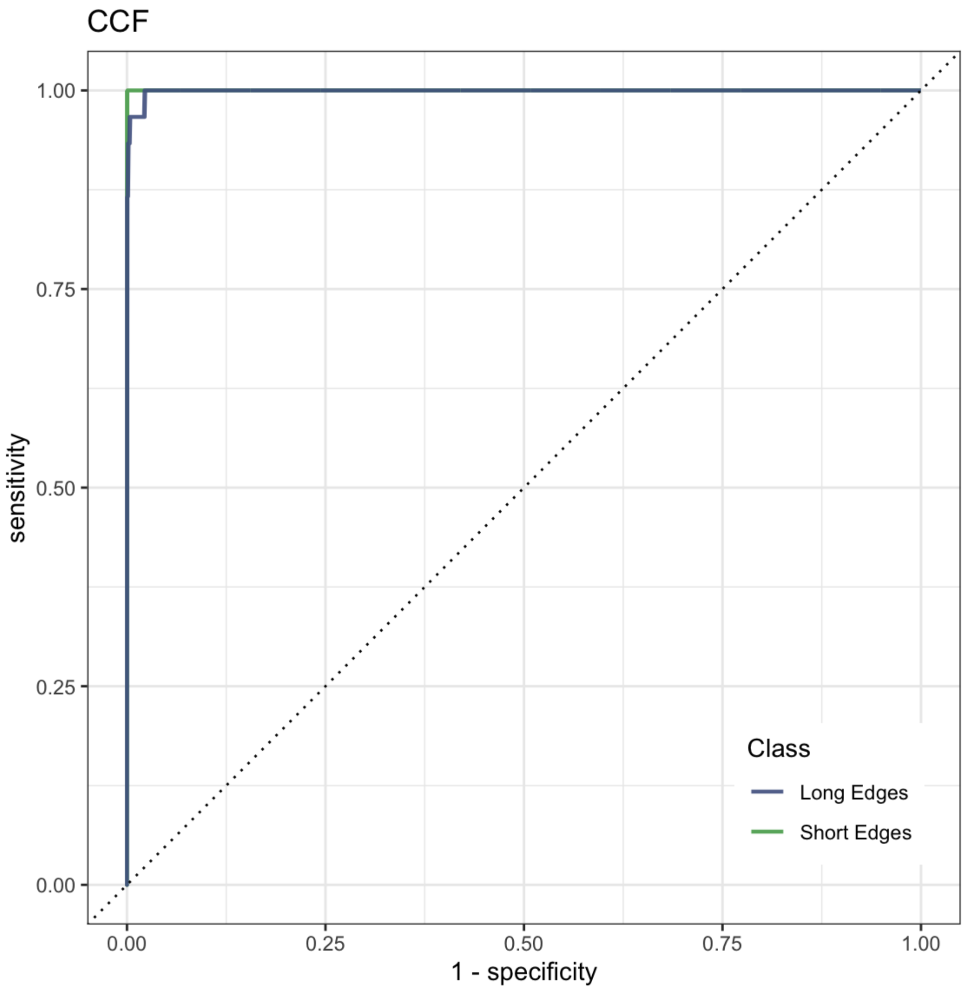

Figure 9. ROC curves for CCF-based classification. Curves are shown separately for long and short edge classes; the dotted diagonal indicates chance-level classification.

Table 6. Comparison of CCF means and standard deviations across different replicates, stagings, and acquisition settings. Dashes indicate that the standard deviation is undefined because only one CCF value is available.

	Replicate (R)		**Staging (S)**			**Acquisition (A)**		
Edge	**Avg**$_R$	**SD**$_R$	**Avg**$_S$	**SD**$_S$	**Avg**$_S$ **-** **Avg**$_R$	**Avg**$_A$	**SD**$_A$	**Avg**$_A$ **-** **Avg**$_S$
A	0.818	0.015	0.961	0.012	0.143	0.992	0.003	0.031
B	0.879	0.031	0.922	0.042	0.043	0.939	—	0.017
C	0.841	0.022	0.952	0.021	0.111	0.959	—	0.007
D	0.797	0.011	0.966	0.017	0.169	0.986	—	0.020

the manual extraction of profiles from scans. To support reproduction of the results reported here, the extraction locations and values for each scan have been included.

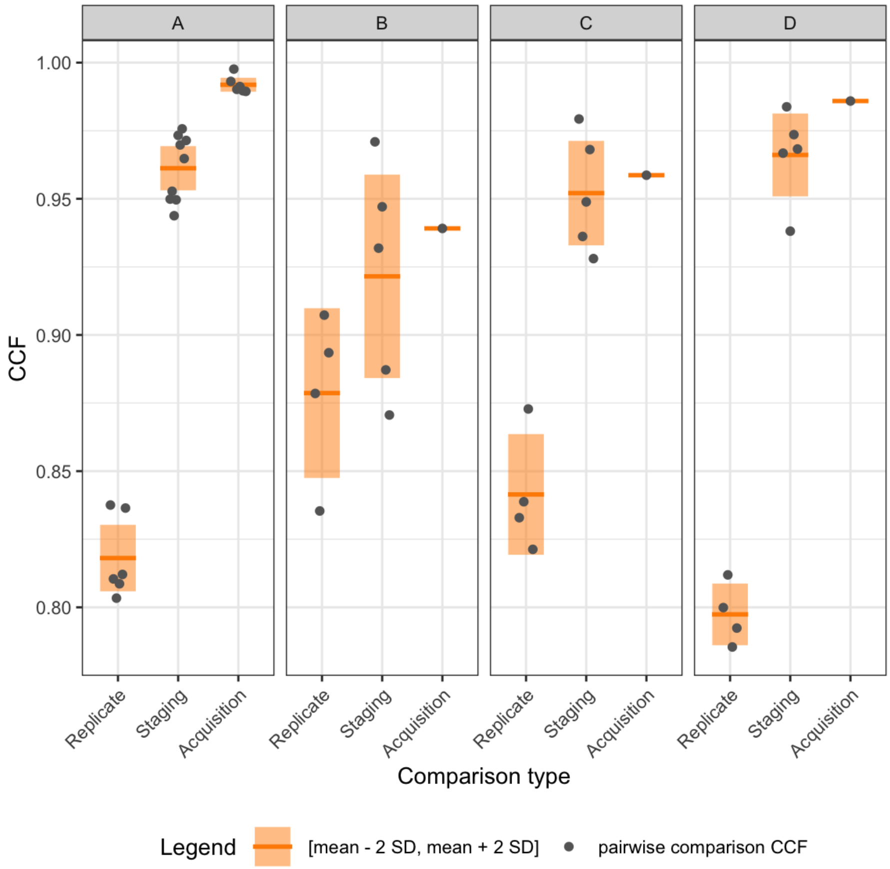

Figure 10. Variability across scanning conditions. Panels A, B, C, and D correspond to the four cut surfaces. Gray points show pairwise CCF values, orange horizontal lines show means, and shaded intervals extend two standard deviations above and below each mean.

Usage Notes

The raw surface scans are provided in `x3p` format. To inspect or process these files in R, use the `x3ptools` package and its reference manual, available from CRAN[21]. The supplied profiles and processed signals provide a starting point for users who do not wish to repeat manual profile extraction. The supplied extraction coordinates allow these derived files to be related back to the corresponding scans. Because each supplied signal is based on a manually selected profile, users assessing sensitivity to profile placement should re-extract profiles from the raw `x3p` scans using alternative documented locations.

Data Availability

The data described in this Data Descriptor are available from Iowa State University DataShare [DATASET CITATION] at `https://doi.org/[DOI]`. [Agent: Replace the dataset citation and DOI placeholders with the same record used in Data Records.]

Code Availability

Custom R code was used to inspect raw scans, extract profiles, derive signals, align signals, and generate the validation outputs. The code is available in the `assessment/code` folder at `https://github.com/heike/wirecuts-data`. [Agent: For permanent access, archive

Table 7. Overview of available code.

Code	Description	Section
Inspect raw scans		
`1-create_pngs_from_x3p.R`	obtain images of `x3p` files in `scans/`	Scan acquisition
Extract profiles		
`2-create_profiles_from_x3p.R`	manually extract profiles from each scan	Signal extraction
`3-create-single-profile-file.R`	combine profile information	Signal extraction
Derive signals		
`4-create_signals_from_profiles.R`	derive signals from each profile	Signal extraction
Compare signals		
`5-create-images.R`	create images for pairwise alignment	Pairwise alignment of signals
`6-align-pairwise.R`	compute pairwise alignment CCF values	Pairwise alignment of signals
`7-all-comparison-results.R`	visualize comparison results	Data Overview and Technical Validation

the exact code release in a DOI-providing repository and cite it if feasible; Scientific Data recommends combining GitHub with a DOI archive.] The files and their roles are described in Table 7.

The provided code reproduces the processing, comparison, validation, and visualization outputs described in Data Acquisition and Processing and Technical Validation. **The workflow was run using R [version] and `x3ptools` [version]. Versions of additional R packages are recorded in [environment file or repository documentation path]. [Agent: Replace the R, `x3ptools`, and environment-record placeholders after checking the analysis environment.]**

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Author Contributions

[Agent: Confirm whether all authors approve conversion of this statement to the CRediT taxonomy before revising the contributor roles.]
Y.L.: Methodology, Software, Validation, Data Curation, Writing: Draft; H.H.: Conceptualization, Methodology, Validation, Writing: Review and Editing; C.M.: Laboratory Supervision; E.A.: Physical Specimen, Scanning; J.S.: Forensic Advice; A.C.: Funding Acquisition.
All authors reviewed the manuscript.

Competing Interests

H.H. is a technical advisor to the Association of Firearm and Toolmark Examiners (AFTE), a fellow of the American Statistical Association (ASA), and a member of the ASA Forensic Science Committee. H.H. has testified as a court witness on behalf of Judge April Neubauer in the New York State Supreme Court, Criminal Term, in New York City. **[Agent: Confirm that the declaration covers every author and replace Alicia's declaration, if any.]**
The authors declare the following competing interests:

Funding

This work was partially funded by the Center for Statistics and Applications in Forensic Evidence (CSAFE) through Cooperative Agreement 70NANB20H019 between NIST and Iowa State University. The agreement includes activities carried out at Carnegie Mellon University, Duke University, the University of California, Irvine, the University of Virginia, West Virginia University, the University of Pennsylvania, Swarthmore College, and the University of Nebraska-Lincoln.